

BPG-09 · BEST PRACTICE GUIDES

Estimate at completion — choosing and defending a method

The formula was never the hard part

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Every estimate at completion is a claim about the work that has not been done yet, and the formula only encodes which claim you are making. This guide runs the four standard methods over one shared data set so the spread is visible, gives the criteria for selecting between them, shows how to test a forecast for credibility before a board does, and sets out the four sentences a defensible forecast has to survive.

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Estimate at completion — choosing and defending a method

In one paragraph. Every estimate at completion is a claim about the work that has not been done yet, and the formula only encodes which claim you are making. This guide runs the four standard methods over one shared data set so the spread is visible, gives the criteria for selecting between them, shows how to test a forecast for credibility before a board does, and sets out the four sentences a defensible forecast has to survive.

Who this is for. Cost engineers, project controls managers and project managers who own a monthly forecast, and the sponsors and portfolio managers who have to decide whether to believe one.

— 1. A forecast is a decision, not an output

Three competent people can take the same control account data, apply three standard methods, and produce three estimates at completion (EAC) that differ by more than ten per cent of the budget. None of them has made an arithmetic error. They have made different assumptions about the remaining work, and the formulae are simply the shorthand in which those assumptions are written down.

This is the point most forecasting practice misses. The question a forecast answers is not "what does the data say" — the data says several things at once — but "what do we believe about the work ahead, and on what grounds". A team that cannot state its assumption in a sentence has not made a forecast. It has run a spreadsheet.

The professional obligation follows directly: **choose the method for a stated reason, disclose the reason, and say what would change it.** A forecast you cannot defend is a guess with decimals.

— 2. The identity underneath every method

Every EAC obeys one identity:

$$EAC = AC + ETC$$

where **AC** is actual cost to date and **ETC** is the estimate to complete — the forecast cost of the work remaining. Nothing else in the family is new. The methods differ only in how ETC is derived, and each derivation is an assumption about how the remaining work will behave relative to the work already done.

Two supporting quantities:

$VAC = BAC - EAC$ Variance at completion; negative = forecast overrun
 $TCPI = (BAC - EV) \div (BAC - AC)$ The cost efficiency the remaining work must achieve to land on BAC

BAC is the budget at completion and EV the earned value; both are defined and measured in BPG-08 – Earned value in practice, which this guide assumes.

— 3. The four methods, and the claim each one makes

Method	ETC derivation	The claim being made
A — Remaining work at budget	$ETC = BAC - EV$	The variance to date came from a discrete, closed event. The rest of the job runs to budget.
B — Remaining work at current cost performance	$ETC = (BAC - EV) \div CPI$, so $EAC = BAC \div CPI$	The inefficiency is systemic and persists at its present rate for the remainder.
C — Remaining work dragged by cost and schedule	$ETC = (BAC - EV) \div (CPI \times SPI)$	Being late is itself inflating cost — extended preliminaries, disruption, acceleration — so the two compound.
D — Bottom-up re-estimate	ETC estimated directly from the remaining scope	Past performance is not representative of what is left, because the remaining work is different in kind.

CPI is the cost performance index and SPI the schedule performance index.

Two remarks that save arguments later.

Method B has two forms that are algebraically identical. $AC + (BAC - EV) \div CPI$ reduces to $BAC \div CPI$ whenever CPI is computed as $EV \div AC$ on cumulative figures, because $AC = EV \div CPI$. If your two forms disagree, you have mixed cumulative and period indices, or you are dividing by a CPI that was rounded before use.

Method C is not "the conservative one". It embeds a specific causal story — that schedule slippage is converting into cost — and it is wrong, not merely cautious, on a project that is behind schedule for reasons that cost nothing extra. Reaching for C because it gives the largest number is as unprofessional as reaching for A because it gives the smallest.

— 4. Criteria for choosing

The method should match the **cause** of the variance, and cause is established by investigation, not by inspecting the index. Four questions, in order:

1. What actually caused the variance to date, and is it closed? If the variance came from a discrete event that has finished — a rework episode now complete, a rate spike now locked by a signed order, a one-off remediation — then the remaining work has no reason to inherit it, and method A is defensible. The test is evidence of closure: an executed contract, a completed scope, a signed-off inspection. "We think it was a one-off" is not closure.

2. Is the inefficiency structural? Productivity shortfalls, rate pressure in a tight labour market, an under-resourced supervision model and poor buildability are all conditions rather than events. They persist until something changes them, and method B is the natural forecast. Ask what specific, funded intervention would break the pattern; if nobody can name one, B is the honest answer.

3. Is lateness costing money? Where the project is behind and the delay is drawing time-related cost — extended site establishment, standing plant, retained supervision, acceleration measures — cost and schedule performance genuinely compound and method C reflects the mechanism. Where the project is behind on float-rich work with no time-related consequence, C overstates.

4. Does the remaining work look like the work done? This is the question that overrides the other three. A project moving from bulk construction into commissioning, from civils into fit-out, or from a first-of-a-kind unit into repeat units has a remainder whose cost behaviour is not predicted by the past at all. Extrapolation methods are then structurally unsuited, whatever their arithmetic, and a bottom-up ETC (method D) is the only sound basis.

A fifth consideration cuts across all four: **the stability of the CPI trend**. A cumulative CPI that has been flat for three or four periods is a settled measurement, and a forecast that assumes it will improve is making a claim that needs evidence. Published research on large programmes has examined how early cumulative cost performance stabilises and how rarely it recovers materially thereafter [\[CONFIRM: citation and date for the cumulative-CPI stability research, before publication\]](#). Whatever the precise finding, the practical rule is defensible on its own: once a CPI has settled, "the variance was atypical" is the claim that carries the burden of proof, not the default.

— 5. Reporting a range, not a point

A single EAC implies a precision the data does not support. Better practice is to report:

- the **selected** EAC and its method;
- the **assumption** it rests on, in one sentence;
- a **range** bounded by the plausible methods, with the assumption at each end;
- what would move the forecast, and by roughly how much.

A range is not evasion. It is a statement about which uncertainty is real and which is resolvable. If the range is wide because nobody has re-estimated the remaining scope, say so — that is a resource decision the sponsor can take. If it is wide because a major commercial position is unresolved, that is a different decision and belongs to a different person.

The **to-complete performance index (TCPI)** is the credibility test that should accompany any forecast that is better than method B. It states the cost efficiency the remaining work must achieve. Comparing it with the efficiency actually achieved so far converts an optimistic forecast into a specific, answerable claim: this forecast requires the team to perform 20 % better for the rest of the project than it has to date — here is why that is possible. If no such answer exists, the forecast is not a forecast.

— 6. Defending it to a board

A board does not want the method; it wants to know whether to act. Four sentences carry a forecast:

1. **The number and its movement.** "The forecast at completion is USD 19.2 million, up 0.4 million on last month."
2. **The method and the assumption in the same breath.** "It is a bottom-up re-estimate of the remaining scope, because the commissioning phase ahead does not resemble the construction behind us."

3. **The credibility test.** "It assumes the remaining work runs about 7 % more efficiently than the work to date, which the recovery plan is built to deliver."
4. **The trigger.** "If the commissioning schedule slips beyond the end of next quarter, extended preliminaries add roughly 0.6 million and the forecast moves to the top of the range."

What must not happen is the reverse order — the method explained first, the number last, the assumption never. And a forecast should never be presented as having "improved" because the method changed. Changing method between periods is legitimate when the cause changed; it is a reporting failure when it is not disclosed. State the previous method, the new method, and the reason, every time.

Automation makes this discipline more important, not less. A tool can compute the whole family in milliseconds and rank the results; it cannot know whether the commissioning phase resembles the civils. AI proposes; the professional disposes — and signs.

— 7. How this goes wrong

One method, applied forever, because it is in the template. The most common failure. $BAC \div CPI$ becomes the house forecast, and the assumption embedded in it is never revisited even when the project changes character entirely.

Method chosen by the answer it gives. Selecting downwards to protect a position, or upwards to build private contingency, is the same defect in opposite directions. Both are visible over time as forecast bias — a project whose EAC only ever moves in one direction was not forecasting, it was conceding.

Rounding the index before dividing by it. On a budget of 18,000,000, a CPI of 0.8949 gives $18,000,000 \div 0.8949 = 20,113,979$. Round the index to 0.89 first and the same calculation gives $18,000,000 \div 0.89 = 20,224,719$ — about 111,000 of forecast created by rounding. Carry full precision through the calculation and round only the presented result.

Forecasting the whole project as one number. EAC computed on project totals hides offsetting movements: a control account 400,000 under and another 900,000 over produce a comfortable-looking aggregate. Forecast at control account level, then aggregate.

Mixing the reserves into the forecast. Contingency drawn down for a materialised risk belongs in the forecast; contingency still held against open risk does not belong in an EAC as though it were spent, and management reserve is outside the baseline altogether. [BPG-10 – Contingency and management reserve](#) sets out the treatment; confusing the two is how a project reports an overrun it does not have, or hides one it does.

A bottom-up ETC that was not actually built bottom-up. "We re-estimated" frequently means the previous number was adjusted by a judgement. A genuine bottom-up ETC has a scope list, quantities, rates and a basis-of-estimate note. Anything else is method A with a different label.

Silent method changes. The forecast improves month on month and the reason is that the method rotated. This destroys the credibility of the whole reporting line, because a reader who spots it once will discount every subsequent forecast.

Ignoring what the forecast implies about the schedule. An EAC that assumes recovery of cost efficiency while the programme assumes the current durations is two documents describing different projects.

— 8. Worked example

Illustrative figures. Currency USD; data date the end of Month 10 of an 18-month baseline; all figures cumulative; indices carried at full precision and presented to four decimal places; a single control account portfolio.

The shared data set.

Quantity	Value
Budget at completion (BAC)	18,000,000
Planned value (PV)	7,680,000
Earned value (EV)	7,200,000
Actual cost (AC), including accruals	8,000,000

Derived:

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CPI      = EV ÷ AC = 7,200,000 ÷ 8,000,000 = 0.9000
SPI      = EV ÷ PV = 7,200,000 ÷ 7,680,000 = 0.9375
CPI × SPI = 0.9000 × 0.9375 = 0.84375
Percent complete = EV ÷ BAC = 7,200,000 ÷ 18,000,000 = 40.0 %
Work remaining at budget = BAC - EV = 18,000,000 - 7,200,000 = 10,800,000
Budget remaining in cash = BAC - AC = 18,000,000 - 8,000,000 = 10,000,000

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Method A — remaining work at budget.

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ETC = BAC - EV = 10,800,000
EAC = AC + ETC = 8,000,000 + 10,800,000 = 18,800,000
VAC = BAC - EAC = 18,000,000 - 18,800,000 = (800,000)

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Method B — remaining work at current cost performance.

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ETC = (BAC - EV) ÷ CPI = 10,800,000 ÷ 0.9000 = 12,000,000
EAC = AC + ETC = 8,000,000 + 12,000,000 = 20,000,000
Cross-check: EAC = BAC ÷ CPI = 18,000,000 ÷ 0.9000 = 20,000,000 ✓
VAC = 18,000,000 - 20,000,000 = (2,000,000)

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Method C — cost and schedule compounding.

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ETC = (BAC - EV) ÷ (CPI × SPI) = 10,800,000 ÷ 0.84375 = 12,800,000
EAC = 8,000,000 + 12,800,000 = 20,800,000
VAC = 18,000,000 - 20,800,000 = (2,800,000)

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Method D — bottom-up re-estimate of the remainder. The remaining scope is re-estimated from quantities and rates rather than extrapolated:

Remaining scope element	ETC
Remaining installation work	6,900,000
Commissioning and handover	2,400,000
Extended site preliminaries for the forecast two-month slip	1,300,000
Escalation on subcontract packages not yet let	600,000
ETC (bottom-up)	11,200,000

$$\begin{aligned} \text{EAC} &= \text{AC} + \text{ETC} = 8,000,000 + 11,200,000 = 19,200,000 \\ \text{VAC} &= 18,000,000 - 19,200,000 = (1,200,000) \end{aligned}$$

The spread, side by side.

Method	Assumption	EAC	VAC
A — remaining work at budget	The variance was a closed, one-off event	18,800,000	(800,000)
D — bottom-up re-estimate	The remainder is different work, priced directly	19,200,000	(1,200,000)
B — current cost performance persists	The inefficiency is structural	20,000,000	(2,000,000)
C — cost and schedule compound	Lateness is converting into cost	20,800,000	(2,800,000)

The four forecasts span $20,800,000 - 18,800,000 = 2,000,000$, which is $2,000,000 \div 18,000,000 = 11.1\%$ of the budget. That spread is not imprecision. It is the range of futures consistent with the same past, and choosing within it is the professional act.

The credibility test.

$$\text{TCPI to BAC} = (\text{BAC} - \text{EV}) \div (\text{BAC} - \text{AC}) = 10,800,000 \div 10,000,000 = 1.0800$$

Landing on budget requires the remaining 60 % of the work to run at a cost efficiency of 1.08 when the first 40 % achieved 0.90 — an improvement of $1.0800 \div 0.9000 = 1.20$, or 20 %, sustained to completion. Absent a named and funded change, that is not a forecast; it is a wish.

Apply the same test to the selected forecast:

$$\begin{aligned} \text{TCPI to EAC (method D)} &= (\text{BAC} - \text{EV}) \div (\text{EAC} - \text{AC}) \\ &= 10,800,000 \div (19,200,000 - 8,000,000) \\ &= 10,800,000 \div 11,200,000 = 0.9643 \end{aligned}$$

Even the bottom-up forecast assumes the remaining work runs at 0.9643 against 0.9000 achieved — an improvement of $0.9643 \div 0.9000 = 1.0714$, about 7 %. That is a much smaller claim than 20 %, but it is still a claim, and it belongs in the narrative. A forecast whose embedded improvement is never stated is optimistic by default.

Selecting, and saying why. The remaining scope includes commissioning and handover — work of a different character from the installation performed to date — so extrapolation methods are structurally unsuited, and method D is chosen. The bottom-up ETC lands at 19,200,000, inside the

range set by the formula methods, which is a reassurance rather than a validation: it means the re-estimate has not drifted away from what performance to date implies.

The board paragraph. "Forecast at completion USD 19.2 million against a budget of 18.0 million, a projected overrun of 1.2 million. The basis is a bottom-up re-estimate of the remaining scope, chosen because the commissioning phase ahead does not resemble the installation work behind us, so performance to date does not predict it. The forecast assumes the remaining work runs about 7 % more cost-efficiently than the work to date, which the revised supervision model is intended to deliver. If that improvement does not materialise, the persisting-performance forecast is 20.0 million; if the two-month slip extends, extended preliminaries take it towards 20.8 million. Recovery to the 18.0 million budget would require a 20 % sustained improvement in cost efficiency and is not credible on current evidence."

Assumptions this example depends on. Earning rules are objective and unchanged; accruals are booked so AC is complete at the data date; no instructed variation is awaiting incorporation into the baseline; the contingency held against open risk is excluded from all four EAC figures and reported separately; escalation is included only in the bottom-up ETC, where it is explicit.

— 9. Checklist

Use this before the forecast leaves your desk.

Before calculating

- [] The cost ledger is closed at the data date and accruals are booked.
- [] Instructed variations awaiting incorporation are identified and quantified separately.
- [] Contingency drawn down is separated from contingency still held.
- [] The cause of the cumulative variance has been investigated, not inferred from the index.

Calculating

- [] The forecast is built at control account level and aggregated, not computed on project totals.
- [] Indices are carried at full precision and rounded only for presentation.
- [] Where method B is used, both forms agree.
- [] Where method D is used, there is a scope list, quantities, rates and a basis-of-estimate note.
- [] Escalation and currency exposure are either in the ETC explicitly or excluded explicitly.

Testing

- [] TCPI to BAC computed, and the required improvement expressed as a percentage of performance to date.
- [] TCPI to the selected EAC computed, so the improvement embedded in your own forecast is visible.
- [] The forecast is consistent with the current schedule — the same slip, the same durations.
- [] A range is stated with the assumption at each end.

Reporting

- [] The method is named, and the assumption stated in one sentence.
- [] Any change of method since last period is disclosed with its reason.
- [] The movement since last period is explained by cause, not by method.

- [] The trigger that would move the forecast is named, with its approximate value.

— Related

- [BPG-08 – Earned value in practice](#) — the measurement discipline every EAC method depends on; read it first if your CPI is not trusted.
- [BPG-10 – Contingency and management reserve](#) — what belongs in the forecast, what sits beside it, and what sits outside the baseline entirely.
- [BPG-16 – Risk registers that work](#) — where the quantified exposure behind a forecast range comes from.
- [BPG-14 – Monthly reporting that gets read](#) — how the forecast and its assumption reach a decision-maker intact.
- [TPL-08 – Estimate at completion scenario comparison](#) — the instrument for running §8 on your own data.

— Sources and standards

The EAC family and the to-complete performance index are described in published earned value frameworks, including the AACE International Total Cost Management framework and the PMBOK Guide; their principles are explained here in our own words and no text or table is reproduced. The internal reference is BoK Domain 6 (EVM/EAC) and BoK Domain 3 (Budgeting & Forecasting). All figures in §8 are illustrative and were computed for this document. One citation remains outstanding, marked in §4.

— Status and version

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